Frequency-Offset Information Aided Self Time Synchronization Scheme for High-Dynamic Multi-UAV Networks

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***Abstract*— Due to the unique merits of unmanned aerial vehicle (UAV) systems, they have already been harnessed for military, public, and civil applications. Time synchronization is a significant premise of formatting and applying UAV networks. However, the irregular high-speed mobile UAVs pose new challenges to time synchronization, especially when external time references are unavailable in some rigid scenarios. Therefore, in these harsh cases, self-time-synchronization (STS) without any external assistance should be concerned to overcome the relative velocity between UAVs caused by irregular high-speed motion. In this paper, a realistic timestamps model for the multi-UAV networks is established, and then a dynamic topology-based maximum likelihood estimator will be developed to carry out the STS. Furthermore, by introducing the information on frequency offset, a new estimator with the closed-form expression is proposed based on a two-way message exchange framework. After that, a tracking algorithm with the assistance of estimation results will be introduced to compensate for the time-varying change of the clock parameters for the dynamic topology UAV networks. To evaluate the performance of the estimator, both the estimation error and Cramér-Rao lower bound are analyzed. Numerical results show that the proposed algorithm exhibits its superiority in STS performance and computational complexity compared to the existing two-way message exchange algorithm using timestamps only.**

***Index Terms*— Multi-UAV networks, dynamic topology, self-time-synchronization, frequency offset, relative radial velocity.**

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I. Introduction

OWING to the merits of convenient deployment, simple structure, good concealment, and landing flexibility, unmanned aerial vehicles (UAVs) are paving the way for a broad range of applications [1] like emergency networks in disasters [2], hyper-dense networks [3] and lost people rescue networks [4]. The multi-UAV networks can break through some limitations of a single UAV while processing the perception, the decision, and the execution [5]. Time synchronization is the fundamental technology for multi-UAV networks, including but not limited to network establishment, data fusion, and resource allocation [6]. Generally, external synchronization, which uses the global navigation satellite system (GNSS) or mobile telecommunication networks as the time benchmark, is the most common time synchronization scheme [7], [8], [9]. However, these external time benchmarks are only sometimes reliable while facing obstacles or harsh weather [10], [11], [12]. Therefore, self-time-synchronization (STS), which can promote time synchronization of the networks without an external time benchmark, is given more prominence [13].

A. The Existing Work

In multi-UAV networks, each UAV has its own independent timescale generated by the local clock oscillator. Since the STS is a procedure of synchronizing each UAV to a unique time benchmark [14], [15]. In this process, the difference in UAVs' timescale will drift from each other over time, which is caused by imperfections in oscillator circuits and hardware aging. Then, the timescale at each UAV should be evaluated, adjusted, and maintained. Therefore, the research on STS mainly emphasizes two aspects: Clock parameters estimation and tracking.

1) The Estimation of Clock Parameters: The most popular clock parameter estimation algorithms for the multi-UAV networks originated from those of wireless sensor networks, which assume that the multi-UAV networks keep stationary during STS [16], [17], [18]. However, this assumption is unrealistic in practical multi-UAV networks, resulting in the serious performance degradation of the STS [19]. To deal with the irregular motions among UAVs in the process of the STS, a variety of algorithms for the STS have been introduced by researchers. For instance, a timestamps model was proposed to characterize the movements among UAVs in [20]. However, the propagation delay between message exchanges has been ignored. A more realistic model was established in [21], where an iterative Gauss-Newton algorithm was adopted to estimate the clock parameters. However, the iterative algorithm requires an accurate initial guess and exhibits a high computational complexity. To tackle this issue, a closed-form expression was described in [22], where an auxiliary variable was accepted to reduce the complexity.

The application scope and intelligence scale of the multi-UAV networks are determined by the performance of the clock parameters estimation [6]. Nevertheless, The time synchronization performance of the above algorithms, which only use the timestamp information, is limited. Referring to the Doppler effect theory, the frequency offset among UAVs are composed of both the clock skew and the Doppler shift caused by the motion among UAVs. Therefore, the estimation performance can be further improved with the assistance of the frequency offset. Thus, there is still a research gap in clock parameter estimation along with the frequency offset.

2) The Tracking of Clock Parameters: The time of the whole network is synchronized after the clock parameters have been estimated and compensated. However, the clock parameters are time-varying along with the temperature and the oscillator's aging. Although the time-varying characters can be compensated by repeating a new estimation process, this procedure will cost unnecessary time overhead and computing complexity [23]. To reduce such extra cost, varying parameter tracking algorithms were proposed [24]. For example, an Expectation-Maximization clock parameter tracking algorithm was proposed based on timestamps in [25]. Furthermore, a timestamp-free clock skew tracking method was proposed in [26]. Based on [26], clock skew and clock offset simultaneously tracking algorithm were proposed in [27].

However, the current research on the tracking algorithm only focuses on stationary networks, and the tracking algorithm for dynamic topology networks, such as multi-UAV networks, still lackes exploration.

B. The Proposed Technological Approaches

To address these issues as mentioned earlier, the technique of the estimation and tracking algorithm of clock parameters may constitute the major concern for multi-UAV networks. In particular, the following two aspects must be given special attention in the following research:

1) The clock parameters estimation assisted by the frequency offset evaluation should be studied to improve the estimation accuracy and reduce the computation complexity for high-speed multi-UAV networks.
2) The clock parameter tracking algorithm for the dynamic topology of multi-UAV networks should be investigated based on the clock parameters estimation results.

In summary, the highly dynamic topology of multi-UAV networks must be appropriately considered during the STS to exploit the performance benefit.

C. Contributions

This paper proposes a new clock parameter estimation and tracking algorithm to deal with the high-speed motion characteristic of multi-UAV networks. Specifically, this work considers a typical multi-UAV network in which a constant speed of the UAVs' relative movement is assumed during the estimation process. Thus the Doppler frequency shift caused by the relative motion is involved. The main contributions of this paper are summarized as follows.

1) A more realistic time synchronization system model in terms of the high-speed motion for the multi-UAV networks is established. A maximum likelihood (ML) estimator for clock parameters is developed based on the proposed model.
2) By introducing the information on frequency offset, a new estimator with the closed-form expression is proposed based on a two-way message exchanges framework. Furthermore, the Cramér-Rao lower bound (CRLB) of the proposed estimator is derived as the performance benchmark. Derivation and simulation results verify the superior performance and computational complexity to the typical two-way message exchange method.
3) To compensate for the time-varying change of the clock parameters, a tracking algorithm with the assistance of estimation results is proposed for the high-dynamic multi-UAV networks.

The remainder of this article is organized as follows. Section II establishes the time synchronization system model for the high-dynamic multi-UAV networks. Then, the ML estimator of the clock parameters aided by the frequency offset is proposed in Section III. The CRLB of the proposed estimator is obtained in Section IV. Furthermore, a time-varying clock parameters tracking method with the assistance of the estimation results is introduced in Section V. The procedure of the proposed STS scheme is also described in V. In addition, numerical results are provided in VI. Finally, Section VII concludes this paper.

Notations: Unless specified otherwise, the lowercase letters, the bold-faced lowercase letters, and the bold-faced uppercase letters denote the scalars, vectors, and matrices. Furthermore, $\mathrm{Var}(\cdot)$, $(\cdot)^{\mathrm{T}}$ and $\mathrm{CRLB}(\cdot)$ represent the operators of variance, transpose and CRLB, respectively. In addition, $U(a, b)$ is a uniform distribution between the interval (a, b).

II. System Model

One of the most critical applications of multi-UAV networks is to search and rescue lost people in depopulated zones, including remote mountainous regions or forested areas, with the GNSS denial of severe weather. In this scenario, as shown in Fig. 1, one reference UAV node and three child nodes form a typical UAV network [2]. The time of the child nodes should be synchronized to that of the reference node (marked as A). Since the STS process is independent between each child node and the reference one. Without loss of generality, we first study the STS procedure between a randomly chosen child node B and the reference node A, which can be extended

TABLE I
LIST OF SYMBOLS

Symbol	Definitaion
$T_{1,n}$	The n-th timestamps
d_0	The initial distance between nodes
N	The number of timestamps exchange
v	The radial velocity between nodes
η_{AB}	The clock skew between nodes A and B
θ_{AB}	The clock offset between nodes A and B
$\tau_{\mathrm{AB},n}$	The fixed transfer delay from A to B
X_n	The random transfer delay from A to B
Y_n	The random transfer delay from B to A
σ_t^2	The variance of the random transfer delay
$\boldsymbol{\theta}$	The vector of the clock parameter
$\mathbf{q}$	The vector of the timestamps
$\mathbf{z}$	The vector of the randomly noise
$\ln f(\boldsymbol{\theta})$	The log-likelihood expression of $\boldsymbol{\theta}$
$\mathbf{W}_t$	The positive-definite diagonal weighting matrix
$F_{1,n}$,$F_{2,n}$	The ratio of frequency offset to the carrier
c	The speed of light
$\Delta F_{1,n}$,$\Delta F_{2,n}$	The frequency offset estimation noise
σ_t^2	The variance of the frequency offset estimation
$\mathbf{f}$	The vector of the frequency offset estimation
$\mathbf{m}$	The vector of the frequency offset estimation noise
$\boldsymbol{\theta}_f$	The estimation vector with the frequency offset
$\boldsymbol{\theta}_t$	The estimation vector with the timestamps
$\mathbf{G}$	The design matrix
$\mathbf{I}$	The Fisher information matrix
$\Delta\boldsymbol{\theta}_f$	The estimation error vector
w	The state transfer coefficien
u_η	The random jitter

to the whole network. For an energy-constrained multi-UVAs network, the reference UAV node is periodically selected from all nodes. The energy consumption among nodes is balanced by this method [28].

The time synchronization model between nodes A and B based on the two-way message exchange mechanism is shown in Fig. 2. We consider that the initial distance between nodes A and B is d_0, and node B moves away from node A at the radial velocity v. According to the two-way message exchange mechanism, the two nodes exchange N rounds of information. In the n-th round, the reference node A sends a synchronization message to node B at time $T_{1,n}$, which is received by the child node B at time $T_{2,n}$. Then node B replies message to node A at time $T_{3,n}$, which contains the timestamps $T_{2,n}$ and $T_{3,n}$. Finally, this message is then received by node A at time $T_{4,n}$. If some messages are lost during transmission, this round of timestamp exchange fails and does not participate in the calculation process. After N rounds of this message exchange process, node A obtains a set of timestamps $\{T_{1,n}, T_{2,n}, T_{3,n}, T_{4,n} | n \in 1, 2, \cdots, N\}$.

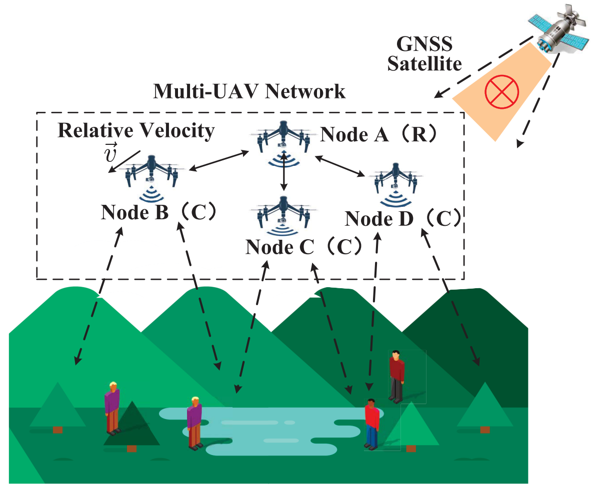

Fig. 1. Multi-UAV network search and rescue for lost people in a remote mountainous region.

Therefore, The procedure as mentioned above can be modeled as

$$\begin{cases} T_{2,n} = \eta_{\mathrm{AB}}(T_{1,n} + \tau_{\mathrm{BA},n} + X_n) + \theta_{\mathrm{AB}}, \\ T_{3,n} = \eta_{\mathrm{AB}}(T_{4,n} - \tau_{\mathrm{AB},n} + Y_n) + \theta_{\mathrm{AB}}, \end{cases} \tag{1}$$

where η_{AB} and θ_{AB} denote the clock skew and the clock offset, respectively. Furthermore, $\tau_{\mathrm{AB},n}$ represents the fixed transfer delay of the n-th round from node A to node B, and $\tau_{\mathrm{BA},n}$ is the fixed transfer delay of the n-th round from node B to node A. In addition, X_n and Y_n are the random parts of the transfer delay, which are assumed to be independent identically distributed (i.i.d) zero-mean Gaussian random variables with a variance of σ_t^2, i.e., $X_n \sim \mathcal{N}(0, \sigma_t^2)$ and $Y_n \sim \mathcal{N}(0, \sigma_t^2)$ [13].

As shown in Fig. 2, since node B is moving away from node A, the transfer of the delay between node A and node B is time-varying. Therefore, the timestamps of the n-th round can be expressed as

$$\begin{cases} T_{2,n} = \eta_{\mathrm{AB}}(T_{1,n} + \tau_{\mathrm{BA},1} + \dfrac{v\Delta T_{1,n}}{(c-v)} + X_n) + \theta_{\mathrm{AB}}, \\ T_{3,n} = \eta_{\mathrm{AB}}(T_{4,n} - \tau_{\mathrm{BA},1} - \dfrac{v\Delta T_{3,n}}{c\eta_{\mathrm{AB}}} - Y_n) + \theta_{\mathrm{AB}}, \end{cases} \tag{2}$$

where $\Delta T_{1,i} = (T_{1,i} - T_{1,1})$ and $\Delta T_{3,i} = (T_{3,i} - T_{2,1})$. If the relative radial velocity v between these two nodes is set to zero, (2) simplifies to Eq (1).

Based on the proposed model in (2), the clock parameters, such as η_{AB} and θ_{AB} should be estimated first to execute the continuing STS process.

III. THE FREQUENCY OFFSET AIDED ESTIMATOR FOR THE CLOCK PARAMETERS

In this section, the typical ML estimator for the proposed model of the multi-UAV networks is obtained as the performance benchmark. After that, a frequency offset-aided estimator is proposed to achieve better performance.

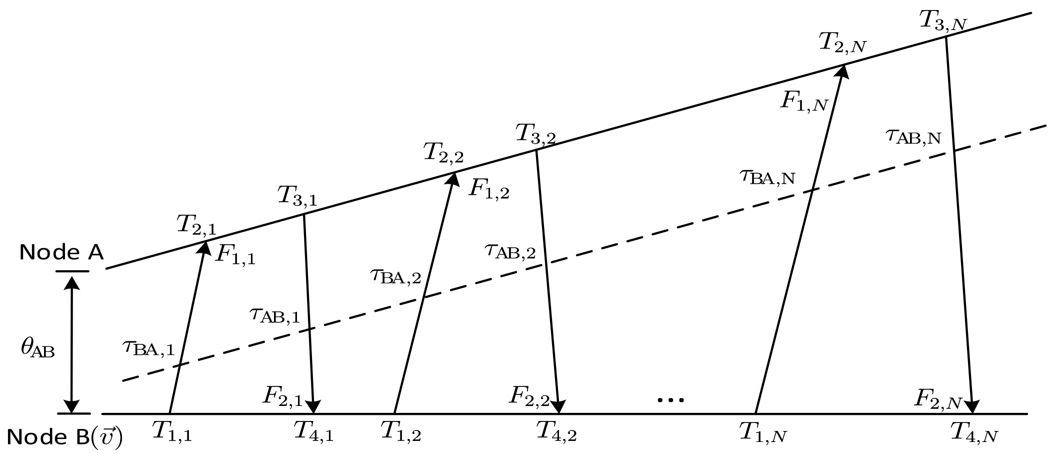

Fig. 2. Two-way Message exchanges model between the reference node A and the child node B.

A. ML Estimator

According to (2), $T_{1,n}$ and $T_{4,n}$ can be derived by

$$\begin{cases} T_{1,n} = \dfrac{T_{2,n}}{\eta_{\mathrm{AB}}} - \tau_{\mathrm{BA},1} - \dfrac{v\Delta T_{1,n}}{c-v} - \dfrac{\theta_{\mathrm{AB}}}{\eta_{\mathrm{AB}}} - X_n, \\ T_{4,n} = \dfrac{T_{3,n}}{\eta_{\mathrm{AB}}} + \tau_{\mathrm{BA},1} + \dfrac{v\Delta T_{3,n}}{c\eta_{\mathrm{AB}}} - \dfrac{\theta_{\mathrm{AB}}}{\eta_{\mathrm{AB}}} + Y_n. \end{cases} \tag{3}$$

The estimator should estimate the clock parameters, including the clock skew η_{AB}, the clock offset θ_{AB}, the initial delay $\tau_{\mathrm{BA},1}$ and the relative radial velocity v. The vector $\boldsymbol{\theta}$ is adopted to denote the set of the parameters mentioned above, which can be described by

$$\boldsymbol{\theta} = \left[\eta_{\mathrm{AB}}, \tau_{\mathrm{BA},1}, v, \theta_{\mathrm{AB}}\right]^{\mathrm{T}}. \tag{4}$$

Based on (4), (3) can be rewritten as

$$\mathbf{q} = \mathbf{h}(\boldsymbol{\theta}) + \mathbf{z}, \tag{5}$$

where $\mathbf{q}$ is the vector of the timestamps, which can be formulated as $\mathbf{q} = [T_{1,1}, T_{4,1}, \cdots, T_{1,N}, T_{4,N}]^{\mathrm{T}}$. $\mathbf{z} = [-X_1, Y_1, \cdots, -X_N, Y_N]^{\mathrm{T}}$ denotes the randomly noise vector. $\mathbf{h}(\cdot)$ is a nonlinear function, n-th element of this function can be formulated by

$$[\mathbf{h}(\boldsymbol{\theta})]_n = \begin{cases} \dfrac{T_{2,n}}{\eta_{\mathrm{AB}}} - \tau_{\mathrm{BA},1} - \dfrac{v\Delta T_{1,n}}{c-v} - \dfrac{\theta_{\mathrm{AB}}}{\eta_{\mathrm{AB}}}, \\ \dfrac{T_{3,n}}{\eta_{\mathrm{AB}}} + \tau_{\mathrm{BA},1} + \dfrac{v\Delta T_{3,n}}{c\eta_{\mathrm{AB}}} - \dfrac{\theta_{\mathrm{AB}}}{\eta_{\mathrm{AB}}}. \end{cases} \tag{6}$$

The log-likelihood expression of $\boldsymbol{\theta}$ can be given as

$$\ln f(\boldsymbol{\theta}) = N\left(\ln\frac{1}{2\pi\sigma_1^2}\right) - \frac{1}{2}||\mathbf{q} - \mathbf{h}(\boldsymbol{\theta})||^2_{\mathbf{W}_t}. \tag{7}$$

The ML estimator of the parameter vector $\boldsymbol{\theta}$ can be considered a weighted least-squares minimization problem, which can be defined as

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} ||\mathbf{q} - \mathbf{h}(\boldsymbol{\theta})||^2_{\mathbf{W}_t}, \tag{8}$$

where $||\mathbf{y}||^2_{\mathbf{W}_t} = \mathbf{y}^{\mathrm{T}}\mathbf{W}_t\mathbf{y}$. $\mathbf{W}_t$ is a positive-definite diagonal weighting matrix, which can be described as

$$\mathbf{W}_t = \mathrm{diag}\underbrace{\left(\frac{1}{\sigma_t^2}, \frac{1}{\sigma_t^2}, \cdots, \frac{1}{\sigma_t^2}, \frac{1}{\sigma_t^2}\right)}_{2\times N}. \tag{9}$$

Based on this dynamic topology-based ML (DTML) estimator defined in (8), the time synchronization of the UAV networks can be achieved by the typical two-way message exchange framework. A DTML-based framework denotes this time synchronization scheme. However, since $\mathbf{h}(\theta)$ is a nonlinear function, the closed-from of the DTML estimator is challenging to derive. To solve this non-linear problem, the frequency offset information should be introduced in the DTML estimator, where the closed-form expression of the estimator can be derived.

B. The Frequency Offset Information Aided DTML Estimator

The dynamic topology of UAV networks arises from the relative velocity among UAV nodes, causing the Doppler effect between every two nodes. The frequency offset is obtained by comparing the frequencies between the received signal and the local oscillator [2]. Specifically, it is obtained through the acquisition and carrier tracking in the physical layer [29], [30]. Then, the Doppler effect combined with the clock skew defines the frequency offset, which can be formulated by

$$\begin{cases} F_{1,n} = \dfrac{v}{c} + \left(\dfrac{1}{\eta_{\mathrm{AB}}} - 1\right) + \Delta F_{1,n}, \\ F_{2,n} = \dfrac{v}{c} - \left(\dfrac{1}{\eta_{\mathrm{AB}}} - 1\right) + \Delta F_{2,n}, \end{cases} \tag{10}$$

where $F_{1,n} = \frac{f_{1,n}}{f_c}$ and $F_{2,n} = \frac{f_{2,n}}{f_c}$. $f_{1,n}$ and $f_{2,n}$ are the frequency offset obtained through the acquisition and carrier tracking. f_c is the carrier frequency. $F_{1,n}$ and $F_{2,n}$ are the ratio of the frequency offset estimation to the carrier frequency f_c respectively at node A or B in the n-th round. Furthermore, $\Delta F_{1,n}$ and $\Delta F_{2,n}$ are the frequency offset estimation noise which satisfies the independent zero-mean Gaussian distribution with the variance of σ_f^2, i.e., $\Delta F_{1,n} \sim \mathcal{N}(0, \sigma_f^2)$ and $\Delta F_{2,n} \sim \mathcal{N}(0, \sigma_f^2)$. Based on (10), the Doppler effect and the clock skew can be separately obtained.

After N rounds of estimation, (10) can be rewritten as

$$\mathbf{f} = \mathbf{B}\boldsymbol{\theta}_f + \mathbf{m}, \tag{11}$$

where $\mathbf{f} = [F_{1,1} + 1, F_{2,1} - 1, \cdots, F_{1,N} + 1, F_{2,N} - 1]^{\mathrm{T}}$, $\mathbf{m} = [\Delta F_{1,1}, \Delta F_{2,1}, \cdots, \Delta F_{1,N}, \Delta F_{2,N}]^{\mathrm{T}}$, and $\boldsymbol{\theta}_f = [v/c, 1/\eta_{\mathrm{AB}}]^{\mathrm{T}}$.

Furthermore, $\mathbf{B}$ can be expressed as

$$\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ \vdots & \vdots \\ 1 & 1 \\ 1 & -1 \end{bmatrix}_{2N\times 2}. \tag{12}$$

Based on (11), $\boldsymbol{\theta}_f$ can be obtained by

$$\hat{\boldsymbol{\theta}}_f = (\mathbf{B}^{\mathrm{T}}\mathbf{B})^{-1}\mathbf{B}^{\mathrm{T}}\mathbf{f}, \tag{13}$$

where $\hat{\eta}_{\mathrm{AB}} = 1/[\hat{\boldsymbol{\theta}}_f]_2$ and $\hat{v} = c[\hat{\boldsymbol{\theta}}_f]_1$. According to (13), the estimation of η_{AB} and v have been obtained as $\hat{\eta}_{\mathrm{AB}}$ and $\hat{v}$.

Therefore, (3) can be further expressed as

$$\underbrace{\begin{bmatrix} T_{1,1} - \dfrac{T_{2,1}}{\eta_{\mathrm{AB}}} + \dfrac{v\Delta T_{1,1}}{c-v} \\ T_{4,1} - \dfrac{T_{3,1}}{\eta_{\mathrm{AB}}} - \dfrac{v\Delta T_{3,1}}{c\eta_{\mathrm{AB}}} \\ \vdots \\ T_{1,N} - \dfrac{T_{2,N}}{\eta_{\mathrm{AB}}} + \dfrac{v\Delta T_{1,N}}{c-v} \\ T_{4,N} - \dfrac{T_{3,N}}{\eta_{\mathrm{AB}}} - \dfrac{v\Delta T_{3,N}}{c\eta_{\mathrm{AB}}} \end{bmatrix}}_{\mathbf{J}} = \underbrace{\begin{bmatrix} -1 & -1 \\ 1 & -1 \\ \vdots & \vdots \\ -1 & -1 \\ 1 & -1 \end{bmatrix}}_{\mathbf{C}} \underbrace{\begin{bmatrix} \tau_{\mathrm{AB},1} \\ \theta_{\mathrm{AB}}/\eta_{\mathrm{AB}} \end{bmatrix}}_{\boldsymbol{\theta}_t} + \underbrace{\begin{bmatrix} X_1 \\ Y_1 \\ \vdots \\ X_N \\ Y_N \end{bmatrix}}_{\mathbf{z}}. \tag{14}$$

The parameters vector $\boldsymbol{\theta_t}$ to be estimated can be derived by a novel estimator named the frequency-offset information-aided DTML (FI-DTML) estimator, which can be expressed as

$$\hat{\boldsymbol{\theta}}_t = (\mathbf{C}^{\mathrm{T}}\mathbf{C})^{-1}\mathbf{C}^{\mathrm{T}}\mathbf{J}, \tag{15}$$

where $\hat{\tau}_{\mathrm{BA}} = [\hat{\boldsymbol{\theta}}_t]_1$ and $\hat{\theta}_{\mathrm{AB}} = \hat{\eta}_{\mathrm{AB}}[\hat{\boldsymbol{\theta}}_t]_2$.

IV. Estimation Performance Analysis

A. The CRLB of DTML

The CRLB for the vector parameter $\boldsymbol{\theta} = [\eta_{\mathrm{AB}}, \tau_{\mathrm{BA},1}, v, \theta_{\mathrm{AB}}]^{\mathrm{T}}$ can be derived by the inverse of the 4×4 Fisher information matrix $\mathbf{I}(\boldsymbol{\theta})$. The log-likelihood function for $\boldsymbol{\theta}$ has already been shown in (7).

By defining $\mathbf{G} = \frac{\partial \mathbf{h}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}$, the n-th round of $\mathbf{G}$ can be calculated by

$$[\mathbf{G}]_{\mathbf{n}} = \begin{bmatrix} D_1 & -1 & -\dfrac{c\Delta T_{1,n}}{(c-v)^2} & -\dfrac{1}{\eta_{\mathrm{AB}}} \\ D_2 & 1 & \dfrac{\Delta T_{3,n}}{c\eta_{\mathrm{AB}}} & -\dfrac{1}{\eta_{\mathrm{AB}}} \end{bmatrix}, \tag{16}$$

where $D_1 = -T_{2,n}/\eta_{\mathrm{AB}}^2 + \theta_{\mathrm{AB}}/\eta_{\mathrm{AB}}^2$ and $D_2 = -T_{3,n}/\eta_{\mathrm{AB}}^2 - v\Delta T_{3,n}/c\eta_{\mathrm{AB}}^2 + \theta_{\mathrm{AB}}/\eta_{\mathrm{AB}}^2$.

Therefore, the Fisher information matrix $\mathbf{I}(\boldsymbol{\theta})$ is given by

$$\mathbf{I}(\boldsymbol{\theta}) = -\mathbb{E}\left[\frac{\partial^2 \ln f(\boldsymbol{\theta})}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}^{\mathrm{T}}}\right] = \mathbf{G}^{\mathrm{T}}\mathbf{W}\mathbf{G}. \tag{17}$$

Based on $\mathbf{I}(\boldsymbol{\theta})$, the CRLB of $\boldsymbol{\theta}$ can be derived by

$$\mathrm{CRLB}(\boldsymbol{\theta}) = \mathbf{I}^{-1}(\boldsymbol{\theta}). \tag{18}$$

B. The Mean Square Error of FI-DTML

1) The MSE of $\boldsymbol{\theta}_f$: We define the estimation error vector as $\Delta\boldsymbol{\theta}_f = \boldsymbol{\theta}_f - \hat{\boldsymbol{\theta}}_f$. Thus the estimation error variance matrix can be expressed as

$$\begin{aligned} \mathrm{Var}\,(\Delta\boldsymbol{\theta}_f) &= \mathrm{Var}[(\mathbf{B}^{\mathrm{T}}\mathbf{B})^{-1}\mathbf{B}^{\mathrm{T}}\mathbf{z}] \\ &= [(\mathbf{B}^{\mathrm{T}}\mathbf{B})^{-1}\mathbf{B}^{\mathrm{T}}]\mathrm{Var}(\mathbf{z})[(\mathbf{B}^{\mathrm{T}}\mathbf{B})^{-1}\mathbf{B}^{\mathrm{T}}]^{\mathrm{T}} \\ &= [(\mathbf{B}^{\mathrm{T}}\mathbf{B})^{-1}\mathbf{B}^{\mathrm{T}}](\mathbf{W}_f)^{-1}[(\mathbf{B}^{\mathrm{T}}\mathbf{B})^{-1}\mathbf{B}^{\mathrm{T}}]^{\mathrm{T}}, \end{aligned} \tag{19}$$

where the diagonal matrix $\mathbf{W}_f$ is expressed by

$$\mathbf{W}_f = \mathrm{diag}\underbrace{\left(\frac{1}{\sigma_f^2}, \frac{1}{\sigma_f^2}, \cdots, \frac{1}{\sigma_f^2}, \frac{1}{\sigma_f^2}\right)}_{2\times N}. \tag{20}$$

Following by (19), the mean square error of $\boldsymbol{\theta}_f$ can be calculated by

$$\begin{aligned} \boldsymbol{\theta}_f^{\mathrm{MSE}} &= \mathbb{E}\left[\left(\boldsymbol{\theta}_f - \hat{\boldsymbol{\theta}}_f\right)\left(\boldsymbol{\theta}_f - \hat{\boldsymbol{\theta}}_f\right)^{\mathrm{T}}\right] \\ &= \mathbb{E}\left[(\Delta\boldsymbol{\theta}_f)(\Delta\boldsymbol{\theta}_f)^{\mathrm{T}}\right] \\ &= \mathrm{tr}(\mathrm{Var}\,(\Delta\boldsymbol{\theta}_f)) + [\mathbb{E}\,(\Delta\boldsymbol{\theta}_f)]^2 \\ &= \mathrm{tr}(\mathrm{Var}\,(\Delta\boldsymbol{\theta}_f)), \end{aligned} \tag{21}$$

where $\mathrm{tr}(\cdot)$ denotes the trace of a matrix.

2) The MSE of $\tau_{BA,1}$ and θ_{AB}: Usually, $v \ll c$ and $|\eta_{\mathrm{AB}}| \ll 1$ are satisfied. Thus $v/(c-v) \approx v/(c\eta_{\mathrm{AB}}) \approx v/c$. By defining that $\theta_1 = 1/\eta_{\mathrm{AB}}$, $\theta_2 = \tau_{\mathrm{BA},1}$, $\theta_3 = v/c$ and $\theta_4 = \theta_{\mathrm{AB}}$. The MSE of the initial delay $\tau_{\mathrm{BA},1}$ and the clock offset θ_{AB} can be expressed as (22) and (23), as shown at the bottom of the next page,

where $\sigma_{G_1}^2 = \sigma_{G_2}^2 = \frac{\sigma_f^2}{4N}$. The detail deviations of (22) and (23) are shown in the Appendix.

C. The CRLB of the FI-DTML Estimator

1) The CRLB of $\boldsymbol{\theta}_f$: The log-likelihood function of the vector $\boldsymbol{\theta}_f$ can be driven by

$$\ln f(\boldsymbol{\theta}_f) = N\left(\ln\frac{1}{2\pi\sigma_f^2}\right) - \frac{1}{2}||\mathbf{f} - \mathbf{B}\boldsymbol{\theta}_f||_{\mathbf{W}_f}^2. \tag{24}$$

Thus, the Fisher matrix can be calculated by

$$\mathbf{I}(\boldsymbol{\theta}_f) = -\mathbb{E}\left[\frac{\partial^2 \ln f(\boldsymbol{\theta}_f)}{\partial\boldsymbol{\theta}_f\partial\boldsymbol{\theta}_f^{\mathrm{T}}}\right] = \mathbf{B}^{\mathrm{T}}\mathbf{W}_f\mathbf{B} = \begin{bmatrix} \dfrac{2N}{\sigma_f^2} & 0 \\ 0 & \dfrac{2N}{\sigma_f^2} \end{bmatrix}. \tag{25}$$

Then, the CRLB of $\boldsymbol{\theta}_f$ can be derived by

$$\mathrm{CRLB}(\boldsymbol{\theta}_f) = \mathbf{I}^{-1}(\boldsymbol{\theta}_f). \tag{26}$$

Assuming $\mathbf{g}(\boldsymbol{\theta_f}) = [v\ \eta_{\mathrm{AB}}]^{\mathrm{T}}$, the Jacobian matrix of $\mathbf{g}(\boldsymbol{\theta_f})$ can be expressed as

$$\frac{\partial \mathbf{g}(\boldsymbol{\theta}_f)}{\partial\boldsymbol{\theta}_f} = \begin{bmatrix} c^2 & 0 \\ 0 & \eta_{\mathrm{AB}}^4 \end{bmatrix}. \tag{27}$$

Obviously, the CRLB of $\mathbf{g}(\boldsymbol{\theta_f})$ can be derived as $\left[\frac{\partial \mathbf{g}(\boldsymbol{\theta}_f)}{\partial \boldsymbol{\theta}_f}\mathrm{I}^{-1}(\boldsymbol{\theta}_f)\frac{\partial \mathbf{g}(\boldsymbol{\theta}_f)}{\partial \boldsymbol{\theta}_f}^T\right]$. Therefore, the CRLB of each parameter in $\mathbf{g}(\boldsymbol{\theta_f})$ can be obtained as

$$\begin{cases} \mathrm{CRLB}(\eta_{\mathrm{AB}}) = \eta_{\mathrm{AB}}^4 \dfrac{\sigma_f^2}{2N}, \\ \mathrm{CRLB}(v) = c^2 \dfrac{\sigma_f^2}{2N}. \end{cases} \tag{28}$$

2) The CRLB of $\tau_{BA,1}$ and $\theta_{A\,B}$: The CRLB of the initial delay $\tau_{\mathrm{BA},1}$ and the clock offset θ_{AB} can be expressed by

$$\begin{cases} \mathrm{CRLB}(\tau_{\mathrm{BA}}) = \tau_{\mathrm{BA},1}^{\mathrm{MSE}}, \\ \mathrm{CRLB}(\theta_{\mathrm{AB}}) = \theta_{\mathrm{AB}}^{\mathrm{MSE}}, \end{cases} \tag{29}$$

where the detail deviation of (29) is shown in the Appendix.

D. The Effect of Time Synchronization Performance on the Overall Performance of the Multi-UAV Network

V. The Time-Varying Clock Skew Tracking Based on Timestamp Sequence

Due to the temperature variation, the resonant frequency aging, and the phase noise of the crystal resonator, the clock skews and offset values are usually slow-time-varying. Furthermore, the relative velocity, affected by the wind strength and speed micro-adjusting, is therefore slow-time-varying fluttering. To compensate for the above-mentioned time-variant characteristic, a frequency-offset aided time-varying tracking algorithm based on the timestamp sequence will be developed.

Considering the clock skew, a typical model [24] to describe the time-varying character between two oscillators of UAVs is the first-order Gaussian Markov model, which can be expressed as

$$\eta[i] = w \cdot \eta[i-1] + u_\eta[i], \tag{30}$$

where $\eta[i]$ is the clock skew of two UAV nodes at the timestamp i. w denotes the state transfer coefficient to reflect the variation trend of two adjacent two timestamps. We define the coefficient w that satisfies a damped exponential model expressed by $w = \rho^{\tau/T}$, where T, τ, and ρ denote a given period used for normalization, the timestamp estimation period, and a positive number close to 1, respectively. Furthermore, $u_\eta[i]$ is the random jitter satisfying the Gaussian distribution with the variance of $\sigma_{u_\eta}^2 = (1-w^2)\sigma_\eta^2$, where σ_η^2 is the variance of $\eta[i]$.

Since the varying tendency of the fluttering relative velocity is always randomly and slightly vibrating, which can be assumed as a portion of the random jitter. The time-varying model involving both the clock skew and the relative velocity can be defined as

$$\underbrace{\begin{bmatrix} v[i] \\ \eta[i] \end{bmatrix}}_{\mathbf{s}[i]} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & w \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} v[i-1] \\ \eta[i-1] \end{bmatrix}}_{\mathbf{s}[i-1]} + \underbrace{\begin{bmatrix} u_v[i] \\ u_\eta[i] \end{bmatrix}}_{\mathbf{u}[i]}, \tag{31}$$

where $u_v[i]$ is the driving noise of the relative velocity satisfying the Gaussian distribution with mean zero and variance $\sigma_{u_v}^2$. Therefore, the covariance matrix of $\mathbf{u}[i]$ can be calculated as $\mathbf{Q} = \begin{bmatrix} \sigma_{u_v}^2 & 0 \\ 0 & \sigma_{u_\eta}^2 \end{bmatrix}$.

According to (3), the estimated timestamp sequence can be rewritten as

$$\underbrace{T_{4,n} - T_{1,n}}_{R_{1,n}} = \frac{1}{\eta_{\mathrm{AB}}} \underbrace{(T_{3,n} - T_{2,n})}_{R_{2,n}} + 2\tau_{\mathrm{BA},1} + \frac{v\Delta T_{3,n}}{c\eta_{\mathrm{AB}}} + \frac{v\Delta T_{1,n}}{c-v} + \underbrace{X_n + Y_n}_{U_n}, \tag{32}$$

where $U_n \sim \mathcal{N}(0, 2\sigma_t^2)$.

Since the information exchange period among nodes is concise, the clock skew and the relative velocity are assumed to remain constant in two rounds. Thus (32) can be further expressed as (33), as shown at the bottom of the next page. According to (33), the frequency offset shown in (10) can be calculated by

$$\begin{cases} \underbrace{F_{1,n} + F_{1,n+1}}_{F_1'[i]} = 2\dfrac{v}{c} + 2\left(\dfrac{1}{\eta_{\mathrm{AB}}} - 1\right) + \underbrace{\Delta F_{1,n} + \Delta F_{1,n+1}}_{\Delta F_1'[i]}, \\ \underbrace{F_{2,n} + F_{2,n+1}}_{F_2'[i]} = 2\dfrac{v}{c} - 2\left(\dfrac{1}{\eta_{\mathrm{AB}}} - 1\right) + \underbrace{\Delta F_{2,n} + \Delta F_{2,n+1}}_{\Delta F_2'[i]}. \end{cases} \tag{34}$$

Combining (32) and (33) and according to the estimated timestamp sequence, the time-varying character model between two oscillators of UAVs can be derived as

$$\underbrace{\begin{bmatrix} F_1'[i] \\ F_2'[i] \\ R_1'[i] \end{bmatrix}}_{\mathbf{x}[i]} = \mathbf{p}(\mathbf{s}[i]) + \underbrace{\begin{bmatrix} \Delta F_1'[i] \\ \Delta F_2'[i] \\ U'[i] \end{bmatrix}}_{\mathbf{y}[i]}, \tag{35}$$

$$\tau_{\mathrm{BA},1}^{\mathrm{MSE}} = \frac{\sigma_{G_1}^2}{4N^2}\left[\sum_{n=1}^{N}(T_{2,n} - T_{3,n} - \Delta T_{1,n} - \Delta T_{3,n})\right]^2 + \frac{\sigma_{G_2}^2}{4N^2}\left[\sum_{n=1}^{N}(T_{3,n} - T_{2,n} - \Delta T_{1,n} - \Delta T_{3,n})\right]^2 + \frac{\sigma_t^2}{2N} \tag{22}$$

$$\theta_{\mathrm{AB}}^{\mathrm{MSE}} = \frac{\sigma_{G_1}^2}{4N^2\hat{\theta}_1^2}\left[\sum_{n=1}^{N}(-\Delta T_{1,n} + T_{2,n} + \Delta T_{3,n} + T_{3,n} - 2\theta_4)\right]^2 + \frac{\sigma_{G_2}^2}{4N^2\hat{\theta}_1^2}\left[\sum_{n=1}^{N}(-\Delta T_{1,n} - T_{2,n} + \Delta T_{3,n} - T_{3,n} + 2\theta_4)\right]^2 + \frac{\sigma_t^2}{2N\hat{\theta}_1^2} \tag{23}$$

where $\mathbf{p}$ can be expressed as

$$\mathbf{p}(\mathbf{s}[i]) = \begin{bmatrix} 2\frac{v}{c} + 2\left(\frac{1}{\eta_{\mathrm{AB}}} - 1\right) \\ 2\frac{v}{c} - 2\left(\frac{1}{\eta_{\mathrm{AB}}} - 1\right) \\ \frac{1}{\eta_{\mathrm{AB}}} R'_2[i] + \frac{v}{c\eta_{\mathrm{AB}}} \Delta T'_3[i] + \frac{v}{c-v} \Delta T'_1[i] \end{bmatrix}. \tag{36}$$

The observation noise variance matrix of (35) can be formulated as

$$\mathbf{C} = \begin{bmatrix} 2\sigma_f^2 & 0 & 0 \\ 0 & 2\sigma_f^2 & 0 \\ 0 & 0 & 4\sigma_t^2 \end{bmatrix}. \tag{37}$$

Since (35) is nonlinear, an expanded Kalman filter, a powerful tool to achieve approximate optimal state tracking of a nonlinear system , can be used to track the clock skew variation and relative velocity variation. This scheme can be described as

1) Prediction: $\hat{\mathbf{s}}[i|i-1] = \mathbf{A}\hat{\mathbf{s}}[i-1|i-1]$.
2) The calculation of the Prediction's MSE: $\mathbf{M}[i|i-1] = \mathbf{A}\mathbf{M}[i-1|i-1]\mathbf{A}^{\mathrm{T}} + \mathbf{Q}$.
3) Kalman Gain Solving: $\mathbf{K}[i] = \frac{\mathbf{M}[i|i-1]\mathbf{P}^{\mathrm{T}}[i]}{(\mathbf{C}+\mathbf{P}[i]\mathbf{M}[i|i-1]\mathbf{P}^{\mathrm{T}}[i])}$.
4) Correction: $\hat{\mathbf{s}}[i|i] = \hat{\mathbf{s}}[i|i-1] + \mathbf{K}[i](\mathbf{x}[i] - \mathbf{p}(\hat{\mathbf{s}}[i|i-1]))$.
5) The MSE of the Solutions: $\mathbf{M}[i|i] = (\mathbf{I} - \mathbf{K}[i]\mathbf{P}[i])\mathbf{M}[i|i-1]$.

where $\mathbf{P}[i]$ denotes the Jacobin matrix of $\mathbf{p}$, which can be expressed as

$$\mathbf{P}[i] = \frac{\partial \mathbf{p}}{\partial \mathbf{s}[i]} = \begin{bmatrix} 2/c & -2/\eta_{\mathrm{AB}}^2 \\ 2/c & 2/\eta_{\mathrm{AB}}^2 \\ D_3 & D_4 \end{bmatrix}\Bigg|_{\mathbf{s}[i]=\hat{\mathbf{s}}[i|i-1]}, \tag{38}$$

where $D_3 = (1/(c\eta_{\mathrm{AB}}))\,\Delta T'_3[i] + \left[c/(c-v)^2\right]\Delta T'_1[i]$ and $D_4 = -\left(1/\eta_{\mathrm{AB}}^2\right) R'_2[i] - \left(v/(c\eta_{\mathrm{AB}}^2)\right)\Delta T'_3[i]$.

The algorithm flow of the proposed frequency-offset information aiding STS (FI-STS) algorithm, which combines the clock parameters estimation and tracking respectively shown in Sections IV and V, is summarized in Algorithm 1.

Algorithm 1 The Frequency-Offset Information Aiding STS Algorithm

Input: frequency offset measurements $F_{1,n}, F_{2,n}$ and timestamps $T_{1,n}, T_{2,n}, T_{3,n}, T_{4,n}$, $n = 1, \cdots, N$, noise variance σ_t^2, σ_f^2, $\sigma_{u_\eta}^2$, and $\sigma_{u_v}^2$, the state transfer coefficient w, maximum tracking count $times$.
Step#0 Initial Clock Skew and Relative Radial Velocity Estimation
$\mathbf{f} \leftarrow [F_{1,1}+1, F_{2,1}-1, F_{1,2}+1, F_{2,2}-1]^{\mathrm{T}}$
$\mathbf{B} \leftarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}^{\mathrm{T}}$
$\hat{\theta}_f \leftarrow (\mathbf{B}^{\mathrm{T}}\mathbf{B})^{-1}\mathbf{B}^{\mathrm{T}}\mathbf{f}$
$\hat{v} \leftarrow c[\hat{\boldsymbol{\theta}}_f]_1$ and $\hat{\eta}_{\mathrm{AB}} \leftarrow 1/[\hat{\boldsymbol{\theta}}_f]_2$
$\hat{\mathbf{s}}[-1|-1] \leftarrow [\hat{v}, \hat{\eta}_{\mathrm{AB}}]^{\mathrm{T}}, \mathbf{M}[-1|-1] \leftarrow \mathrm{diag}([\sigma_{u_v}^2, \sigma_{u_\eta}^2])$

Step#1 Clock Skew and Relative Radial Velocity Tracking
for $i = 1 : times$ **do**
Prediction: $\hat{\mathbf{s}}[i|i-1] \leftarrow \mathbf{A}\hat{\mathbf{s}}[i-1|i-1]$
Prediction MSE: $\mathbf{M}[i|i-1] \leftarrow \mathbf{A}\mathbf{M}[i-1|i-1]\mathbf{A}^{\mathrm{T}} + \mathbf{Q}$

Kalman Gain: $\mathbf{K}[i] \leftarrow \frac{\mathbf{M}[i|i-1]\mathbf{P}^{\mathrm{T}}[i]}{(\mathbf{C}+\mathbf{P}[i]\mathbf{M}[i|i-1]\mathbf{P}^{\mathrm{T}}[i])}$
Correction: $\hat{\mathbf{s}}[i|i] \leftarrow \hat{\mathbf{s}}[i|i-1] + \mathbf{K}[i](\mathbf{x}[i] - \mathbf{p}(\hat{\mathbf{s}}[i|i-1]))$

MSE: $\mathbf{M}[i|i] \leftarrow (\mathbf{I} - \mathbf{K}[i]\mathbf{H}[i])\mathbf{M}[i|i-1]$
$\hat{v} \leftarrow \hat{\mathbf{s}}[i|i]_1$ and $\hat{\eta}_{\mathrm{AB}} \leftarrow \hat{\mathbf{s}}[i|i]_2$
$\mathbf{C} \leftarrow \begin{bmatrix} -1 & 1 & -1 & 1 \\ -1 & -1 & -1 & -1 \end{bmatrix}^{\mathrm{T}}$
$\hat{\boldsymbol{\theta}}_t \leftarrow (\mathbf{C}^{\mathrm{T}}\mathbf{C})^{-1}\mathbf{C}^{\mathrm{T}}\mathbf{j}$
$\hat{\tau}_{\mathrm{BA}} \leftarrow [\hat{\boldsymbol{\theta}}_t]_1$ and $\hat{\theta}_{\mathrm{AB}} \leftarrow \hat{\eta}_{\mathrm{AB}}[\hat{\boldsymbol{\theta}}_t]_2$
end
Output: $\hat{\mathbf{s}}[i|i], \hat{\tau}_{\mathrm{BA}}, \hat{\theta}_{\mathrm{AB}}$

According to (13) and (15), the clock offset is calculated by matrix multiplications and inversions, thus the operation times of the multiplication and the addition are adopted to mirror the computational complexity. The computational complexity of the clock offset estimation by the proposed algorithm FI-DTML is shown in Table. II, where that of the existing algorithm two-way TOA localization and synchronization (TWLAS) [21] are used for comparison. N denotes the message exchange process rounds, and L_i is the iteration number of the TWLAS. Obviously, the cost of the computational resource has been significantly reduced by adopting the FI-DTML algorithm, which confirms the superior performance of the complexity.

VI. Numerical Analysis

In this section, the performance of the proposed frequency offset aiding the STS algorithm will be evaluated in terms of different factors. The performance enhancement of the proposed FI-DTML scheme can be proved by comparing it

$$\underbrace{R_{1,n+1} - R_{1,n}}_{R'_1[i]} = \frac{1}{\eta_{\mathrm{AB}}} \underbrace{(R_{2,n+1} - R_{2,n})}_{R'_2[i]} + \frac{v}{c\eta_{\mathrm{AB}}} \underbrace{(\Delta T_{3,n+1} - \Delta T_{3,n})}_{\Delta T'_3[i]} + \frac{v}{c-v} \underbrace{(\Delta T_{1,n+1} - \Delta T_{1,i})}_{\Delta T'_1[i]} + \underbrace{U_{n+1} - U_n}_{U'[i]} \tag{33}$$

TABLE II
THE ESTIMATION ALGORITHM'S COMPUTATIONAL COMPLEXITY COMPARISON

Estimator	Parameters	Number of $+/-$	Number of $\times/\div$
FI-DTML	clock offset	$38N$-11	$46N$+6
TWLAS	clock offset	L_i($64N$+28)	L_i($72N$+164)

TABLE III
SIMULATION PARAMETERS OF THE PROPOSED ANALYSIS

Parameters	value
θ_{AB}	$U(-3,3)$ s
η_{AB}	$U(-10,10)$ ppm
τ_{BA}	$U(-1,1)$ s
v	$U(-300,300)$ m/s
d_0	100 km
L	100000

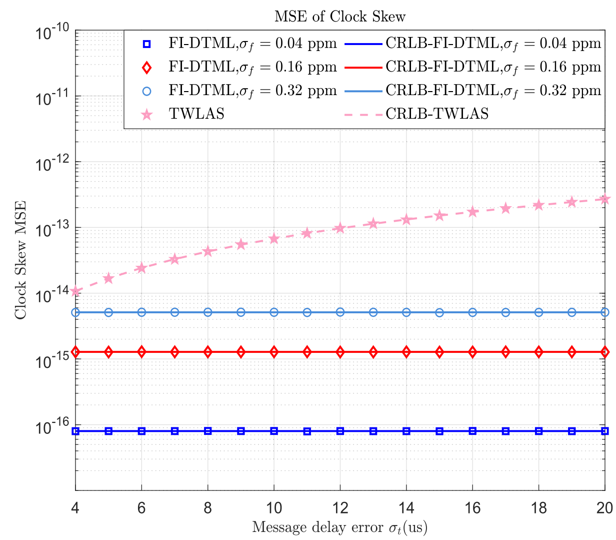

Fig. 3. MSE of the clock skew v.s. the random transfer delay standard deviation σ_t, with the various frequency offset estimation noise σ_f.

with the existing optimal estimation algorithm, which is named TWLAS [21].

A. Simulation Setup

According to the analysis in Section IV, the performance of the proposed algorithm is influenced by several factors, such as the rounds of the message exchange process N, the random transfer delay, and the performance of the frequency offset estimation. In this section, the MSE of the estimation results is adopted as the performance criteria to evaluate the proposed STS algorithm, which is defined as

$$\boldsymbol{\theta}^{\text{MSE}} = \frac{1}{L}\sum_{1}^{L}(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^{\text{T}}, \tag{39}$$

where $\boldsymbol{\theta}$ denotes the parameters to be estimated and $\hat{\boldsymbol{\theta}}$ is the estimation results vector. Furthermore, L is the repeating times of the estimation process.

Without loss of generality, both the initial value of each UAV node's clock offset and clock skew is assumed to follow the uniform distribution $\theta_{\text{A B}} \sim U(-3,3)$s and $\eta_{\text{A B}} \sim U(-10,10)$ppm, which are set randomly at each simulation. The initial distance between nodes A and B is 100 km. The initial delay and relative radial velocity are drawn from uniform distributed $\tau_{\text{B A}} \sim U(-1,1)$s and $v \sim U(-300,300)$m/s. In the following, at each step in the Algorithm. 1, Monte-Carlo simulation will repeat 100000 times to enhance confidence. Furthermore, according to the estimation process of the typical two-way message exchange framework, the TWLAS estimator, which is the baseline scheme, has the same CRLB as the DTML shown in (26).

B. The Estimation Performance in Terms of the Random Transfer Delay

The estimation Performance of the proposed FI-DTML in terms of both the random transfer delay standard deviation σ_t and the frequency offset estimation noise σ_f is plotted in Figs. 3 and 4, respectively, where the TWLAS estimator serves as the performance benchmark. In this section, the standard deviation of the random transfer delay (σ_t) is randomly valued in $[4, 20]$us. The rounds of the message exchange process N is 10.

In Fig. 3, the MSE curves of the clock skew's estimation results, which are the function of σ_t, are described, while σ_f varies. As the frequency offset estimation noise σ_f increases, it is shown that the clock skew estimation performance of FI-DTML decreases gradually. Furthermore, the clock skew MSE of the TWLAS estimator increases while the σ_t increases. However, the σ_t does not affect the proposed FI-DTML estimator. In addition, the simulation results of the proposed FI-DTML estimator nearly reach its CRLB and are superior to the TWLAS, while the standard deviation σ_f is slight enough.

In Fig. 4, the MSE curves of the clock offset's estimation, which is the function of σ_t, are plotted, while the standard deviation σ_f is considered. On the one hand, the clock offset the MSE of both the TWLAS and FI-DTML increase gradually with the rise of σ_t. On the other hand, the MSE curves of FI-DTML slightly decrease while σ_f at first tends to be tender. The influence of σ_f on the clock offset is gradually weakened than the clock skew shown in Fig. 3. Furthermore, the performance of the FI-DTML estimator is superior to the TWLAS.

C. The Estimation Performance in Terms of the Message Exchange Process Rounds

1) The Influence of the Random Transfer Delay: The MSE curves of the clock skew and clock offset estimation, which is the function of the message exchange process rounds N with various message transfer delay estimation noise, are shown in Figs. 5 and 6, respectively, where a fixed frequency offset estimation noise ($\sigma_f = 0.1$ppm) is assumed. During the simulation, the relative radial velocity is uniformly distributed in the interval $[-3e^2, 3e^2]$.

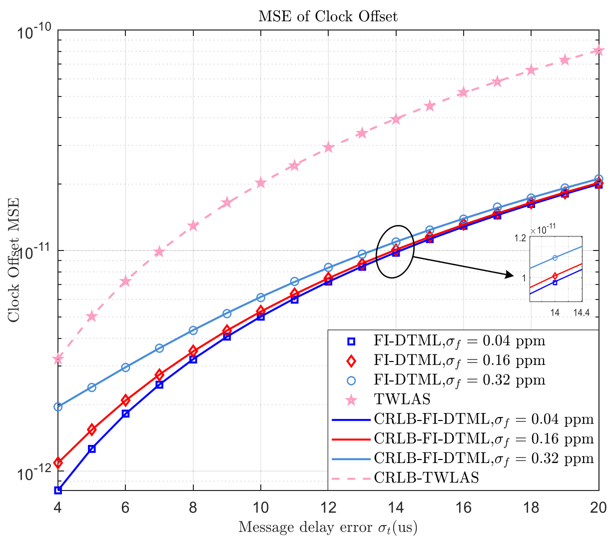

Fig. 4. MSE of the clock offset v.s. the random transfer delay standard deviation σ_t, with the various frequency offset estimation noise σ_f.

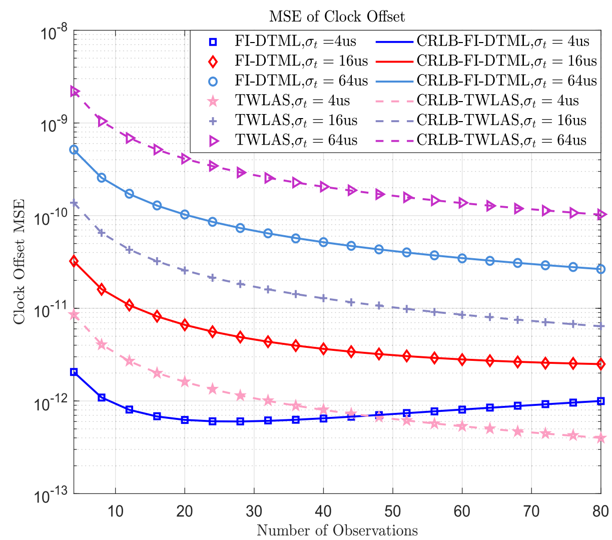

Fig. 6. MSE of the clock offset v.s. the message exchange process rounds N with the various σ_t, where the fixed $\sigma_f = 0.1$ppm is assumed.

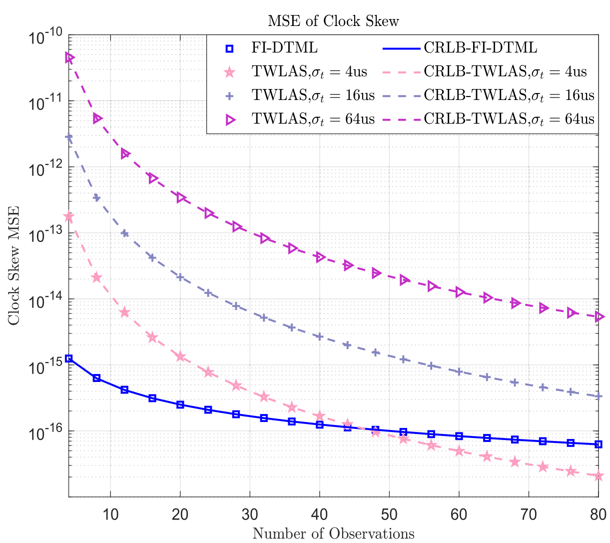

Fig. 5. MSE of the clock skew v.s. the message exchange process rounds N with the various σ_t, where the fixed $\sigma_f = 0.1$ppm is assumed.

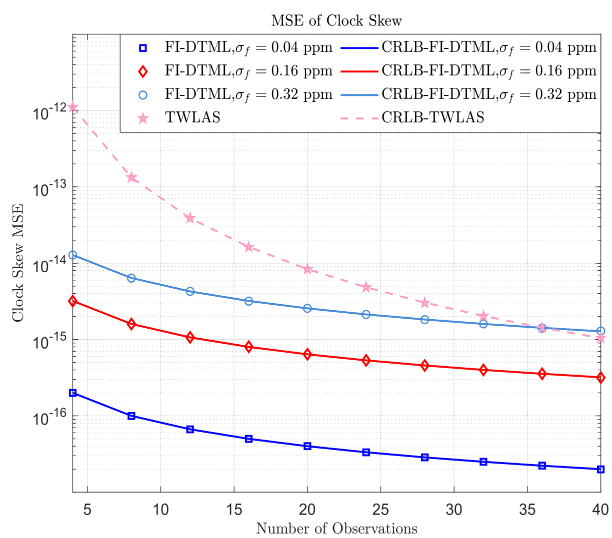

Fig. 7. MSE of the clock skew v.s. the message exchange process rounds N with the various σ_f, where the fixed $\sigma_t = 10$us is assumed.

According to Fig. 5, variant estimators' clock skew estimation performance is a monotonically decreasing function of N (i.e., slowing decline and converging to a floor as N increases). Furthermore, as σ_t decreases, the estimation performance of the TWLAS estimator has been enhanced. In particular, in some specific scenarios (i.e., $N > 50$, $\sigma_t < 4us$, and $\sigma_f = 0.1$ppm), TWLAS estimator outperforms the FI-DTML.

According to Fig. 6, similar to the clock skew, the clock offset estimation performance of variant estimators is a decreasing function of N. In particular, as N increases, the MSE of FI-DTML will creep down at first, but after beyond the breakpoint (i.e., $N = 20$), it will gradually increase. This reveals the fact that the clock offset estimation cannot be enhanced by simply increasing the message exchange process rounds N. Compared with TWLAS, the FI-DTML shows superior performance while $\sigma_t > 4$ or $N < 45$.

2) The Influence of the Frequency Offset Estimation Noise: The MSE curves of the clock skew and clock offset estimation, which are the function of the message exchange process rounds N, are plotted in Figs. 7 and 8, respectively, while a fixed message transfer delay estimation noise ($\sigma_t = 10$us) is assumed. During the simulation, the relative radial velocity is uniformly distributed in the interval $[-3e^2, 3e^2]$.

In Fig. 7, the clock skew estimation results show a performance reduction as the σ_f rises. Therefore, if σ_f is small enough, better performance can be achieved by the proposed FI-DTML algorithm. In Fig. 8, the clock offset estimation performance of FI-DTML descends along with the ascending of σ_f. Compared with the TWLAS estimator, the proposed FI-DTML performs better while σ_f is small enough. In particular, the MSE of the FI-DTML becomes a concave function of N (i.e., $\sigma_f = 0.32$ppm). This observation can be explained as follows. According to (23), the clock offset estimation performance depends on σ_t, σ_f and N. Therefore, with the

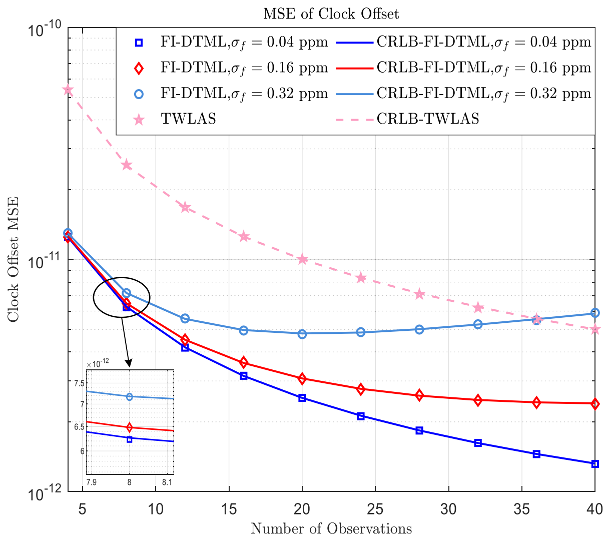

Fig. 8. MSE of the clock offset v.s. the message exchange process rounds N with the various σ_f, where the fixed $\sigma_t = 10$us is assumed.

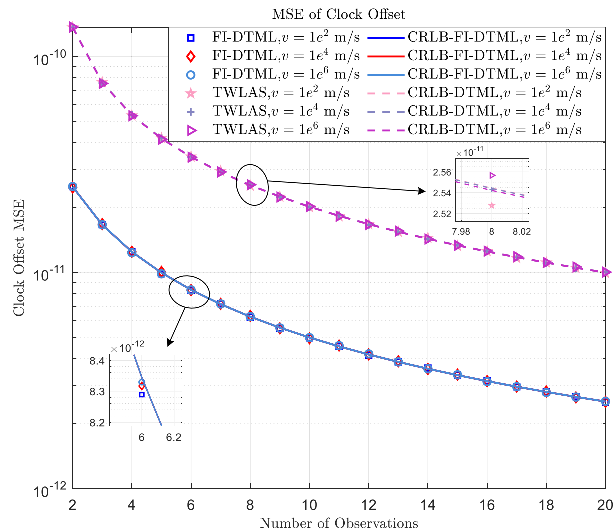

Fig. 10. MSE of the clock offset v.s. the message exchange process rounds N with the various relative radial velocity v.

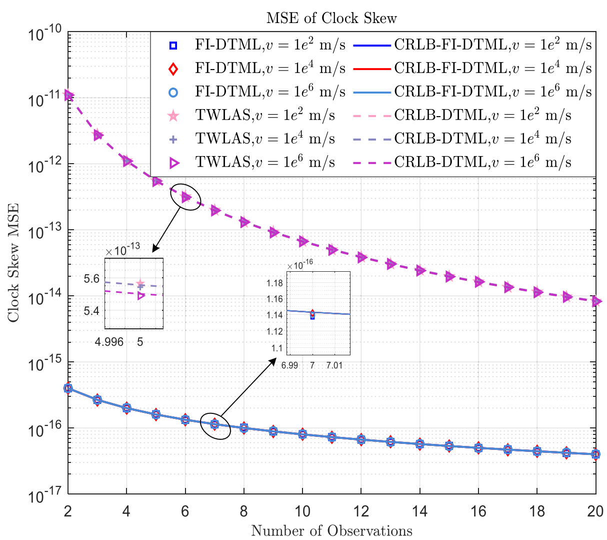

Fig. 9. MSE of the clock skew v.s. the message exchange process rounds N with the various relative radial velocity v.

fixed σ_f, the function degenerates to a concave one or a monotone function with fixed σ_t.

3) The Influence of the Relative Radial Velocity: The MSE curves of the clock skew and clock offset estimation, which are the function of the message exchange process round N with various velocities, are presented in Figs. 9 and 10, respectively, while the fixed message transfer delay estimation noise ($\sigma_t = 10$us) and frequency offset estimation noise ($\sigma_f = 0.04$ppm) are assumed.

As described in Figs. 9 and 10, the MSE curves of both the FI-DTML and the TWLAS algorithm are a monotonically decreasing function of N, that is, firstly decline and then gradually converge to constant. Meanwhile, compared with the TWLAS, the FI-DTML shows tremendous performance benefits. Furthermore, the relative radial velocity plays no role in the decision on the clock skew estimation performance.

TABLE IV
SIMULATION PARAMETERS OF THE PROPOSED ANALYSIS

Parameters	value
Initial relative velocity	200 m/s
Initial clock skew	$20e^{-6}$
τ	2
T	3600 s
ρ	$1 - 2e^{-6}$
σ_t	10us
σ_f	0.1ppm
σ_{u_v}	3(m/s)
σ_{u_η}	0.045ppm

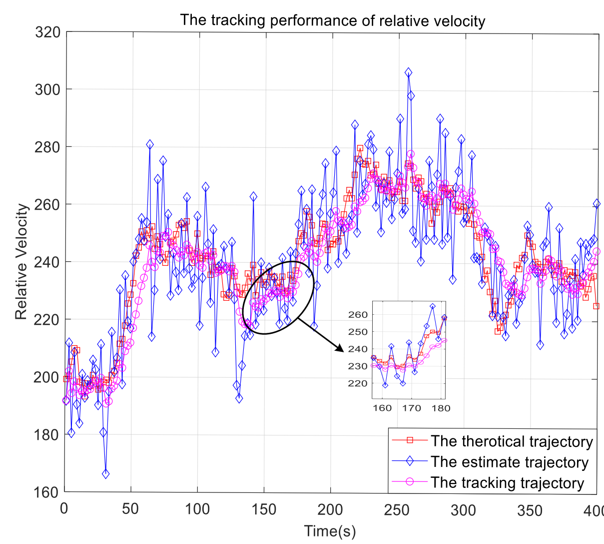

Fig. 11. The relative velocity tracking performance.

D. The Tracking Performance of FI-STS

In this subsection, the state of FI-STS is initialized as $\hat{\mathrm{s}}[-1|-1] = [200\ 20e^{-6}]^{\mathrm{T}}$ and $\mathrm{M}[-1|-1] = \mathrm{diag}([\sigma_{u_v}^2, \sigma_{u_\eta}^2])$.

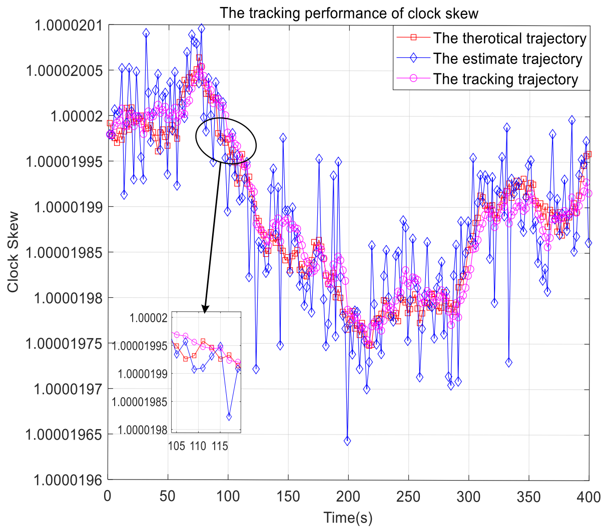

Fig. 12. The clock skew tracking performance.

The initial relative velocity and the clock skew are 200 m/s and $20e^{-6}$ Furthermore, the normalization period $T = 3600$, the timestamp estimation period $\tau = 2$ and the positive number $\rho = 1 - 2e^{-6}$. The clock skew and relative radial velocity tracking the performance of the frequency-offset information aiding STS algorithm have been respectively shown in Figs. 11 and 12, while $\sigma_t = 10$ us, $\sigma_f = 0.1$ ppm, $\sigma_{u_v} = 3$ (m/s), and $\sigma_{u_\eta} = 0.045$ ppm.

The proposed FI-STS algorithm can track the variation tendency of both the clock skew between two oscillators (Fig. 11) and the relative radial velocity between two UAVs (Fig. 12). Furthermore, the estimated time sequence exists in sharp jitter caused by the FI-DTML estimator, which can be significantly reduced to smoothness by the FI-STS.

VII. Conclusion

In this paper, a more realistic time synchronization system model in terms of high-speed motion for multi-UAV networks is established. A maximum likelihood estimator for clock parameters is developed based on the proposed model. By introducing the information of frequency offset, a new estimator with the closed-form expression was presented based on a two-way message exchange framework. Furthermore, the Cramér-Rao lower bound of the proposed estimator has been derived as the performance benchmark. To compensate for the time-varying change of the clock parameters, a tracking algorithm with estimation results was proposed for high-dynamic UAV networks. Numerical results showed that, compared with the DTML and TWLAS, the STS performance can be significantly enhanced by employing the proposed FI-DTML estimator-based STS algorithm. Meanwhile, the performance of the FI-DTML estimator was determined by the message exchange process rounds, the random transfer delay information, and the frequency offset information. Therefore, improving the clock offset estimation performance is impossible by simply increasing the message exchange process rounds. Therefore, by enhancing the accuracy of both the message transfer delay and the Doppler frequency offset, optimal STS performance can be attained.

Appendix A
The Mean Square Error Derivation of $\tau_{\mathrm{BA},1}$ and θ_{AB}

By defining $\theta_1 = 1/\eta_{\mathrm{AB}}$, $\theta_2 = \tau_{\mathrm{BA},1}$, $\theta_3 = v/c$, and $\theta_4 = \theta_{\mathrm{AB}}$, $\boldsymbol{\theta}_t = [\tau_{\mathrm{BA},1}\ \theta_{\mathrm{AB}}/\eta_{\mathrm{AB}}]$ can be expressed as $[\theta_2\ \theta_1\theta_4]$.

According to (14), the estimation of both θ_1 and θ_3 can be expressed as

$$\hat{\theta}_1 = \theta_1 + \left(\underbrace{\frac{1}{2N}\sum_{n=1}^{N}\Delta F_{1,n}}_{G_1} - \underbrace{\frac{1}{2N}\sum_{n=1}^{N}\Delta F_{2,n}}_{G_2}\right) \tag{40}$$

and

$$\hat{\theta}_3 = \theta_3 + \left(\underbrace{\frac{1}{2N}\sum_{n=1}^{N}\Delta F_{1,n}}_{G_1} + \underbrace{\frac{1}{2N}\sum_{n=1}^{N}\Delta F_{2,n}}_{G_2}\right), \tag{41}$$

respectively, where the frequency offset estimation noise $\{\Delta F_{1,n}\}_1^N$ and $\{\Delta F_{2,n}\}_1^N$ are independent identically distributed (i.i.d) zero-mean Gaussian random variables with the variance of σ_f^2.

Therefore, G_1 and G_2 are also zero-mean Gaussian random variables with the variance of $\sigma_f^2/(4N)$ (i.e., $G_1 \sim \mathcal{N}(0, \sigma_f^2/(4N))$ and $G_2 \sim \mathcal{N}(0, \sigma_f^2/(4N))$).

Assuming $v/(c-v) \approx v/(c\eta_{\mathrm{AB}}) \approx v/c$, (3) can be rewritten as

$$\begin{cases} T_{1,n} = \theta_1 T_{2,n} - \theta_2 - \theta_3 \Delta T_{1,n} - \theta_1\theta_4 - X_n, \\ T_{4,n} = \theta_1 T_{3,n} + \theta_2 + \theta_3 \Delta T_{3,n} - \theta_1\theta_4 + Y_n. \end{cases} \tag{42}$$

Substituting (40) and (41) into (42), (42) can be rewritten as (43), shown at the bottom of the next page. According to (15), θ_2 can be estimated

$$\hat{\theta}_2 = \frac{O_{\theta_2}}{2N} = \theta_2 + \underbrace{\frac{1}{2N}\left[\tilde{G}_{1,\theta_2} + \tilde{G}_{2,\theta_2} + \sum_{n=1}^{N}(X_n + Y_n)\right]}_{\tilde{G}_{3,\theta_2}}, \tag{44}$$

where

$$\begin{cases} O_{\theta_2} = \sum_{n=1}^{N}(-T_{1,n} + T_{4,n}) \\ \qquad + \sum_{n=1}^{N}\left[\hat{\theta}_1(T_{2,n} - T_{3,n}) - \hat{\theta}_3(\Delta T_{1,n} + \Delta T_{3,n})\right], \\ \tilde{G}_{1,\theta_2} = \sum_{n=1}^{N}(T_{2,n} - \Delta T_{1,n} - \Delta T_{3,n} - T_{3,n})\, G_1, \\ \tilde{G}_{2,\theta_2} = \sum_{n=1}^{N}(T_{3,n} - \Delta T_{1,n} - T_{2,n} - \Delta T_{3,n})\, G_2, \end{cases} \tag{45}$$

$\tilde{G}_{3,\theta_2}$ is a zero-mean Gaussian random variable, the variance of which can be expressed as (46), shown at the bottom of the page.

Thus,

$$\begin{aligned}\theta_2^{\text{MSE}} &= \mathbb{E}\left(\theta_2-\hat{\theta}_2\right)^2=\mathbb{E}\left(\Delta\theta_2\right)^2=\text{Var}\left(\Delta\theta_2\right)+\left[\text{E}\left(\Delta\theta_2\right)\right]^2\\ &=\frac{\sigma_{G_1}^2}{4N^2}\left[\sum_{n=1}^{N}(T_{2,n}-\Delta T_{1,n}-\Delta T_{3,n}-T_{3,n})\right]^2\\ &\quad+\frac{\sigma_{G_2}^2}{4N^2}\left[\sum_{n=1}^{N}(T_{3,n}-\Delta T_{1,n}-T_{2,n}-\Delta T_{3,n})\right]^2\\ &\quad+\frac{\sigma_t^2}{2N}.\end{aligned}\tag{47}$$

Similar to the θ_2, θ_4 can be expressed as

$$\begin{aligned}\hat{\theta}_4&=\frac{O_{\theta_4}}{2N\hat{\theta}_1}\\ &=\theta_4+\frac{1}{2N\hat{\theta}_1}\underbrace{\left[\tilde{G}_{1,\theta_4}+\tilde{G}_{2,\theta_4}+\sum_{n=1}^{N}(X_n-Y_n)\right]}_{\tilde{G}_{3,\theta_4}},\end{aligned}\tag{48}$$

where

$$\begin{cases}O_{\theta_4}=\sum_{n=1}^{N}(-T_{1,n}-T_{4,n})\\ \qquad+\sum_{n=1}^{N}\left[\hat{\theta}_1(T_{2,n}+T_{3,n})-\hat{\theta}_3(\Delta T_{1,n}-\Delta T_{3,n})\right],\\ \tilde{G}_{1,\theta_4}=\sum_{n=1}^{N}(-\Delta T_{1,n}+T_{2,n}+\Delta T_{3,n}+T_{3,n}-2\theta_4)G_1,\\ \tilde{G}_{2,\theta_4}=\sum_{n=1}^{N}(-\Delta T_{1,n}-T_{2,n}+\Delta T_{3,n}-T_{3,n}+2\theta_4)\,G_2.\end{cases}\tag{49}$$

The variance of $\tilde{G}_{3,\theta_4}$ can then be derived as (50), shown at the top of the next page.

Therefore, the MSE of $\theta_{\text{AB}}^{\text{MSE}}$ can be formulated by

$$\begin{aligned}&\theta_{\text{AB}}^{\text{MSE}}\\ &=\mathbb{E}\left(\theta_4-\hat{\theta}_4\right)^2=\mathbb{E}\left(\Delta\theta_4\right)^2=\text{Var}\left(\Delta\theta_4\right)+\left[\mathbb{E}\left(\Delta\theta_4\right)\right]^2\\ &=\frac{\sigma_{G_1}^2}{4N^2\hat{\theta}_1^2}\left[\sum_{n=1}^{N}(-\Delta T_{1,n}+T_{2,n}+\Delta T_{3,n}+T_{3,n}-2\theta_4)\right]^2\\ &\quad+\frac{\sigma_{G_2}^2}{4N^2\hat{\theta}_1^2}\left[\sum_{n=1}^{N}(-\Delta T_{1,n}-T_{2,n}+\Delta T_{3,n}-T_{3,n}+2\theta_4)\right]^2\\ &\quad+\frac{\sigma_t^2}{2N\hat{\theta}_1^2}.\end{aligned}\tag{51}$$

Appendix B
The CRLB of $\tau_{\text{BA},1}$ and θ_{AB}

The log-likelihood function of $\tau_{\text{BA},1}$ can be written as

$$\ln f(\theta_2)=\ln\frac{1}{\sqrt{2\pi\sigma_{\tilde{G}_{3,\theta_2}}^2}}-\frac{1}{2\sigma_{\tilde{G}_{3,\theta_2}}^2}\left(O_{\theta_2}-2N\theta_2\right)^2.\tag{52}$$

Hence, the CRLB of $\hat{\tau}_{\text{BA},1}$ can be formulated by

$$\begin{aligned}\text{CRLB}(\tau_{\text{BA},1})&=\text{CRLB}(\theta_2)\\ &=-\mathbb{E}\left[\left(\frac{\partial^2\ln f(\theta_2)}{\partial^2\theta_2}\right)\right]^{-1}=\frac{\sigma_{\tilde{G}_{3,\theta_2}}^2}{4N^2}.\end{aligned}\tag{53}$$

Similarly, the log-likihood function of θ_{AB} can be written as

$$\ln f(\theta_4)=\ln\frac{1}{\sqrt{2\pi\sigma_{\tilde{G}_{3,\theta_4}}^2}}-\frac{1}{2\sigma_{\tilde{G}_{3,\theta_4}}^2}\left(O_{\theta_4}-2N\hat{\theta}_1\theta_4\right)^2.\tag{54}$$

the CRLB of $\hat{\theta}_{\text{AB}}$ can be derived by

$$\begin{aligned}\text{CRLB}(\theta_{\text{AB}})&=\text{CRLB}(\theta_4)\\ &=-\mathbb{E}\left[\left(\frac{\partial^2\ln f(\theta_4)}{\partial^2\theta_4}\right)\right]^{-1}\\ &=-\mathbb{E}\left[\frac{\partial\left(\frac{4N\hat{\theta}_1}{2\sigma_{\tilde{G}_{3,\theta_4}}^2}\left(O_{\theta_4}-2N\hat{\theta}_1\theta_4\right)\right)}{\partial\theta_4}\right]^{-1}\\ &=\frac{\sigma_{\tilde{G}_{3,\theta_4}}^2}{4N^2\hat{\theta}_1^2}.\end{aligned}\tag{55}$$

$$\begin{cases}T_{1,n}=\hat{\theta}_1T_{2,n}-\theta_2-\hat{\theta}_3\Delta T_{1,n}-\hat{\theta}_1\theta_4+\left(\Delta T_{1,n}-T_{2,n}+\theta_4\right)G_1+\left(\Delta T_{1,n}+T_{2,n}-\theta_4\right)G_2-X_n,\\ T_{4,n}=\hat{\theta}_1T_{3,n}+\theta_2+\hat{\theta}_3\Delta T_{3,n}-\hat{\theta}_1\theta_4-\left(\Delta T_{3,n}+T_{3,n}-\theta_4\right)G_1-\left(\Delta T_{3,n}-T_{3,n}+\theta_4\right)G_2+Y_n\end{cases}\tag{43}$$

$$\sigma_{\tilde{G}_{3,\theta_2}}^2=\sigma_{G_1}^2\left[\sum_{n=1}^{N}(T_{2,n}-\Delta T_{1,n}-\Delta T_{3,n}-T_{3,n})\right]^2+\sigma_{G_2}^2\left[\sum_{n=1}^{N}(T_{3,n}-\Delta T_{1,n}-T_{2,n}-\Delta T_{3,n})\right]^2+2N\sigma_t^2\tag{46}$$

$$\sigma^2_{\tilde{G}_{3,\theta_4}} = \sigma^2_{G_1}\left[\sum_{n=1}^{N}(-\Delta T_{1,n} + T_{2,n} + \Delta T_{3,n} + T_{3,n} - 2\theta_4)\right]^2 + \sigma^2_{G_2}\left[\sum_{n=1}^{N}(-\Delta T_{1,n} - T_{2,n} + \Delta T_{3,n} - T_{3,n} + 2\theta_4)\right]^2 + 2N\sigma_t^2 \tag{50}$$

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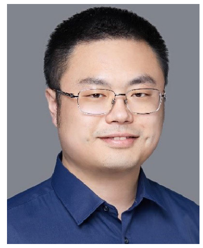

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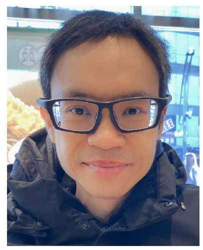

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